

PART 3: COMBINATIONAL LOGIC

DECODERS & ENCODERS

SAAD BAIG – DIGITAL LOGIC DESIGN



The animations and contents in these slides are originally adapted from the DLD course slides designed by Dr. Hasan Baig and Mr. Junaid Ahmed Memon. The original slides can be accessed from <https://www.hasanbaig.com/resources>

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DECODER



- A decoder is a combinational circuit that converts binary information from n input lines to a maximum of 2^n unique output lines.
 - If the n -bit coded information has unused combinations, the decoder may have fewer than 2^n outputs.
 - The name decoder is also used in conjunction with other code converters such as a BCD-to-seven-segment decoder.
- n -to- m line decoder: Inputs: n . Outputs: $m \leq 2^n$
- Any combinational circuit with n inputs and m outputs can be implemented with an n -to- m -line decoder

3-TO-8 LINE DECODER

Truth Table of a Three-to-Eight-Line Decoder

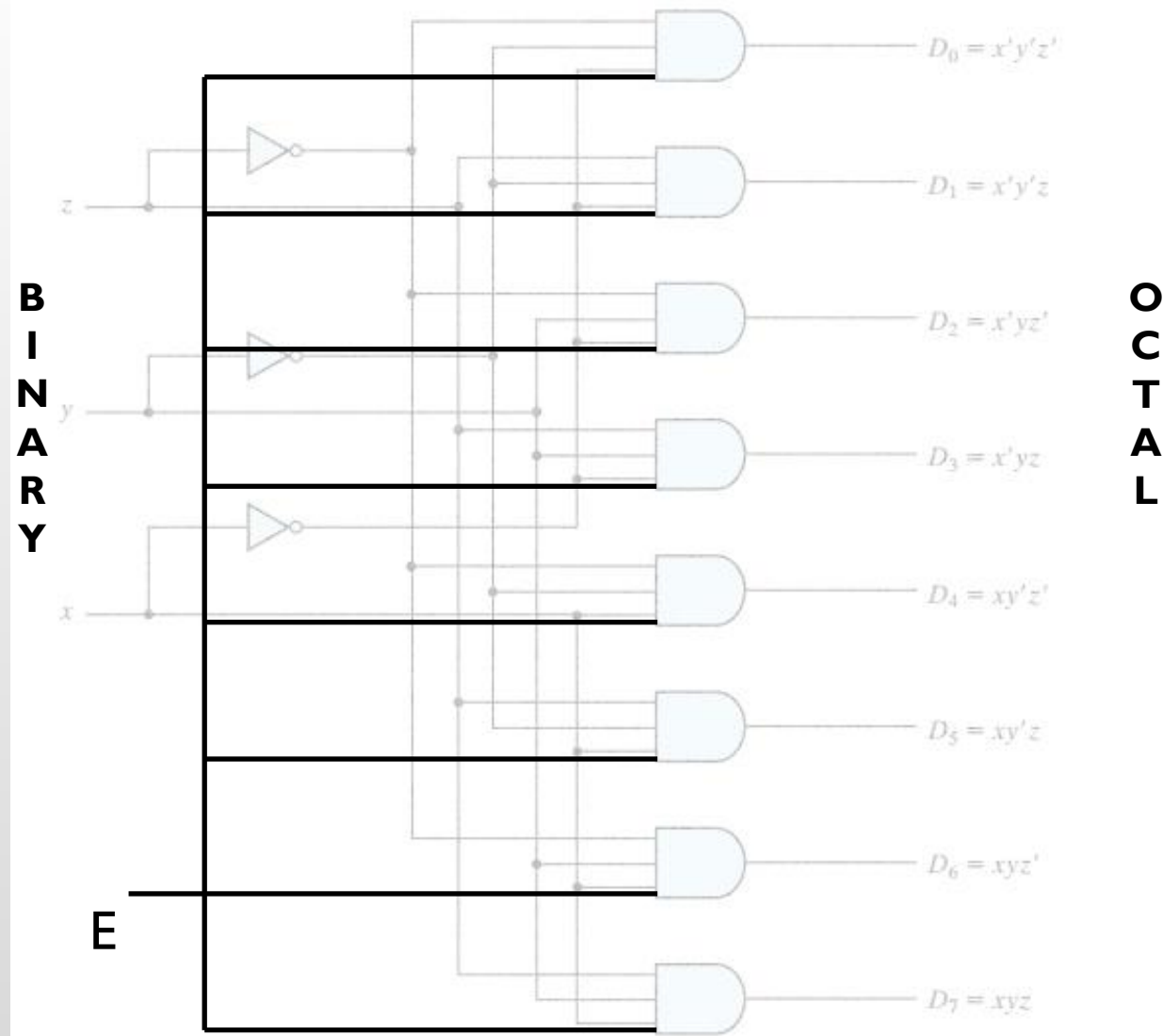
Inputs			Outputs							
x	y	z	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

3-TO-8 LINE DECODER

Application?

A three-to-eight-line decoder can be used for decoding *any* three-bit code to provide eight outputs, one for each element of the code.

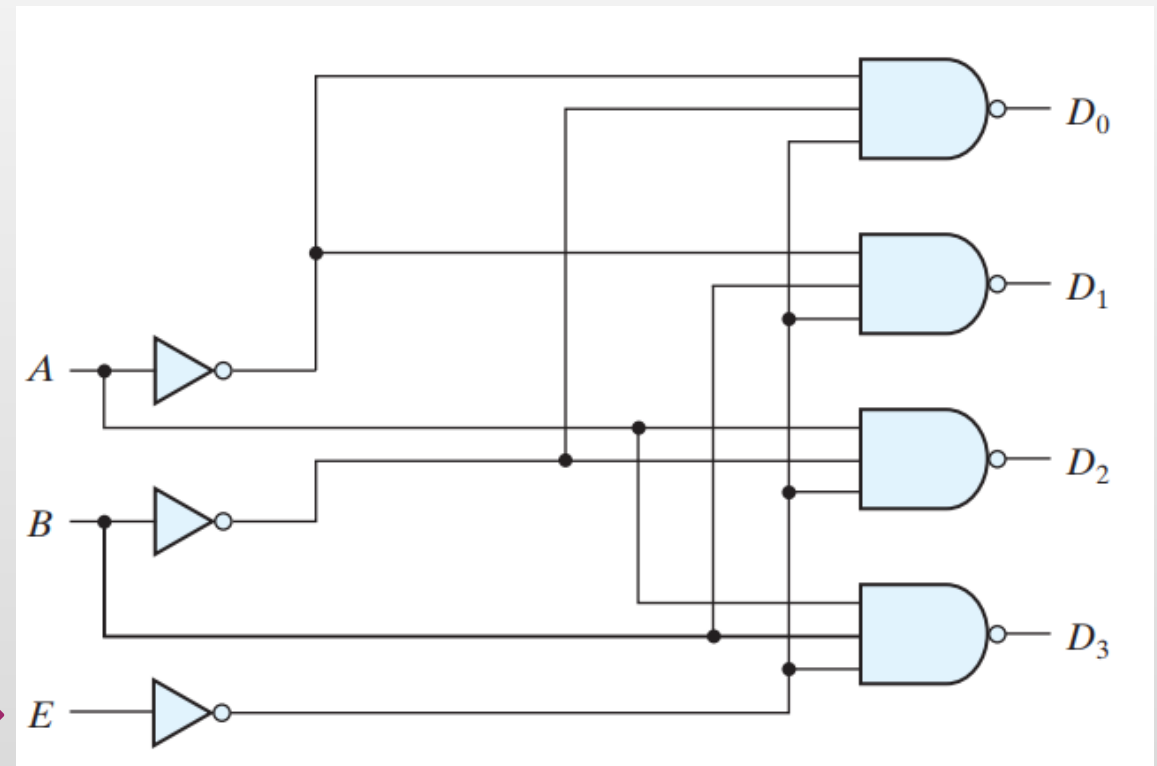
Active High Enable →



2-TO-4 LINE DECODER (USING NAND GATES)

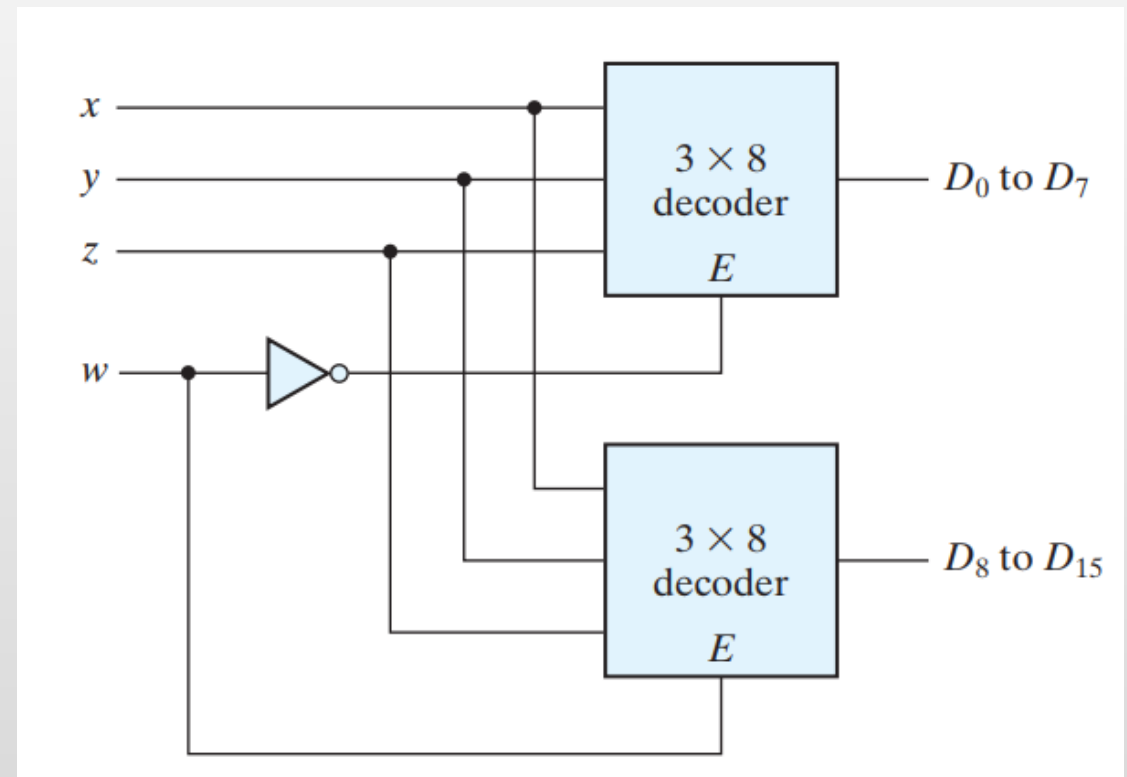
E	A	B	D_0	D_1	D_2	D_3
1	X	X	1	1	1	1
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	1	1	1	0

Active Low Enable →



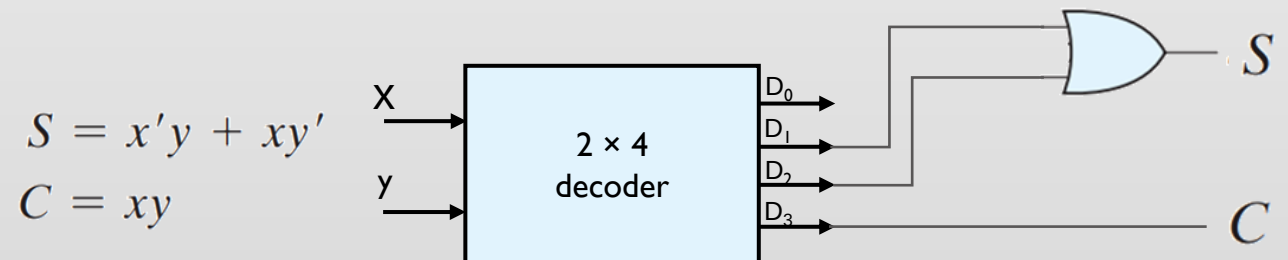
EXTENDING DECODER

- **Example:** 4 x 16 decoder using two 3 x 8 decoders.
- Decoders with enable inputs can be connected together to form a larger decoder circuit.
- In general, enable inputs are a convenient feature for interconnecting two or more standard components for the purpose of combining them into a similar function with more inputs and outputs.



COMBINATIONAL LOGIC IMPLEMENTATION

- A decoder provides the 2^n minterms of n input variables.
- Any Boolean function can be expressed in sum-of-minterms form.
- A decoder that generates the minterms of the function, together with an external OR gate that forms their logical sum, provides a hardware implementation of the function.
 - Example Half-Adder can be implemented using 2×4 decoder.

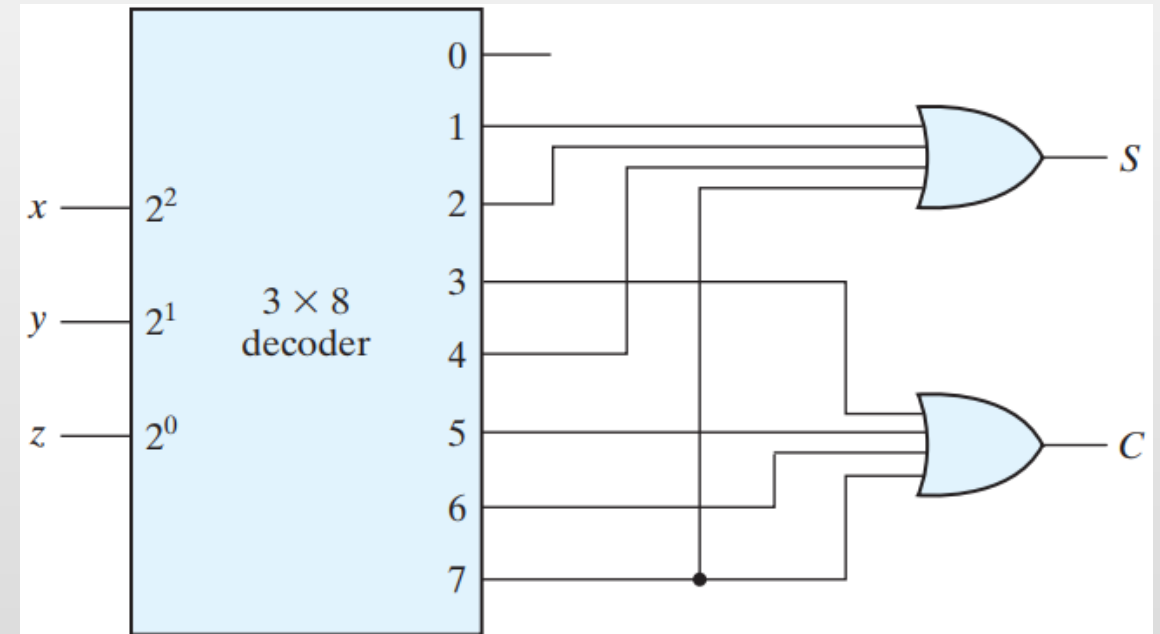


COMBINATIONAL LOGIC IMPLEMENTATION



$$S = x'y'z + x'yz' + xy'z' + xyz$$

$$C = xy + xz + yz$$





DECODERS DESIGN TIPS

- A function with a long list of minterms requires an OR gate with a large number of inputs.
- A function having a list of k minterms can be expressed in its complemented form F' with $2^n - k$ minterms.
- If the number of minterms in the function is greater than $2^n/2$, then F can be expressed with fewer minterms. In such a case, it is advantageous to use a NOR gate to sum the minterms of F' . The output of the NOR gate complements this sum and generates the normal output F .
- If NAND gates are used for the decoder, then the external gates must be NAND gates instead of OR gates. This is because a two-level NAND gate circuit implements a sum-of-minterms function and is equivalent to a two-level AND–OR circuit.

ENCODERS

- An encoder is a digital circuit that performs the inverse operation of a decoder.
- An encoder has 2^n (or fewer) input lines and n output lines.
- Encoder can be implemented with OR gates.
 - **Example:** Octal to Binary Encoder

Truth Table of an Octal-to-Binary Encoder

Inputs								Outputs		
D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	x	y	z
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

$$\begin{array}{l}
 z = D_1 + D_3 + D_5 + D_7 \\
 y = D_2 + D_3 + D_6 + D_7 \\
 x = D_4 + D_5 + D_6 + D_7
 \end{array}$$

$\xrightarrow{8}$
 $\xrightarrow{3}$

PROBLEMS WITH SIMPLE ENCODER

- If two inputs are active simultaneously, the output produces an undefined combination
- Output with all 0's is generated when all inputs are zero but this output is same as when D_0 is equal to 1.

Truth Table of an Octal-to-Binary Encoder

Inputs								Outputs		
D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	x	y	z
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

$$\begin{aligned}
 z &= D_1 + D_3 + D_5 + D_7 \\
 y &= D_2 + D_3 + D_6 + D_7 \\
 x &= D_4 + D_5 + D_6 + D_7
 \end{aligned}$$

PRIORITY ENCODER

- A priority encoder is an encoder circuit that includes the **priority function**.
- The operation of the priority encoder is such that if two or more inputs are equal to 1 at the same time, the input having the highest priority will take precedence.

Truth Table of a Priority Encoder

Inputs				Outputs		
D_0	D_1	D_2	D_3	x	y	V
0	0	0	0	X	X	0
1	0	0	0	0	0	1
X	1	0	0	0	1	1
X	X	1	0	1	0	1
X	X	X	1	1	1	1

Validity Check
Output

ACTIVITY

Priority Encoder



- Draw K-map of below truth table and find minimized Boolean expression of x and y .

Truth Table of a Priority Encoder

Inputs				Outputs		
D_0	D_1	D_2	D_3	x	y	V
0	0	0	0	X	X	0
1	0	0	0	0	0	1
X	1	0	0	0	1	1
X	X	1	0	1	0	1
X	X	X	1	1	1	1

ACTIVITY

Priority Encoder

D_0D_1		D_2D_3			
		00	01	11	10
D_0	00	m_0 X	m_1 1	m_3 1	m_2 1
	01	m_4	m_5 1	m_7 1	m_6 1
	11	m_{12}	m_{13} 1	m_{15} 1	m_{14} 1
	10	m_8	m_9 1	m_{11} 1	m_{10} X

$x = D_2 + D_3$

D_0D_1		D_2D_3			
		00	01	11	10
D_0	00	m_0 X	m_1 1	m_3 1	m_2
	01	m_4 1	m_5 1	m_7 1	m_6
	11	m_{12} 1	m_{13} 1	m_{15} 1	m_{14}
	10	m_8	m_9 1	m_{11} 1	m_{10}

$y = D_3 + D_1D'_2$

ACTIVITY

Priority Encoder



$$\begin{aligned}x &= D_2 + D_3 \\y &= D_3 + D_1 D_2' \\V &= D_0 + D_1 + D_2 + D_3\end{aligned}$$

